

Entangled Gazes: Reconstituting the Stereoscopic Box Set with LLMs and Virtual Reality

Paul Heinrich Bethge*
HKBU

Kowloon Tong, Hong Kong
bethge@hkbu.edu.hk

Adriano Viegas Milani
EPFL

Lausanne, Switzerland
adriano.viegasmilani@epfl.ch

Emmanuelle Denove*
EPFL

Lausanne, Switzerland
emmadeno@protonmail.com

Tai Leong Cheong
CityU

Kowloon Tong, Hong Kong
tlcheong2@cityu.edu.hk

Dhruva Gowda-Storz*
EPFL

Lausanne, Switzerland
dhruva.gowdastorz@epfl.ch

Paul Bourke
Independent Researcher
Melbourne, Australia
paul.bourke@gmail.com

Sarah Kenderdine
EPFL
Lausanne, Switzerland
sarah.kenderdine@epfl.ch

Jeffrey Shaw
HKBU
Kowloon Tong, Hong Kong
jeffreysaw@hkbu.edu.hk

Abstract

This paper presents a computational framework for critically reinterpreting colonial-era stereoscopic box sets. Using James Ricalton's China through the Stereoscope (1901) as a case study, we deconstruct its linear narrative using NLP methods for thematic modeling, semantic mapping, and colonial language identification. This analysis informs two complementary artworks in a large-scale 360° VR environment. Cross Eyed offers a guided experience, using the thematic models as lenses for inquiry and visually distinguishing identified rhetoric. Latent Cartographies, in contrast, empowers users to freely navigate the archive's raw semantic space, fostering diverse modes of experiential engagement with contested heritage. For demonstration visit: <https://epfl-emplus.github.io/EntangledGazes/>

CCS Concepts

• **Applied computing** → **Arts and humanities**; • **Computing methodologies** → *Natural language processing*; • **Human-centered computing** → *Virtual reality*.

Keywords

Archival Art, Stereoscopic Photography, NLP Methods for Art, Virtual Reality

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* Authors contributed equally to this research.



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1 Stereoscopic Box Sets

At the turn of the 20th century, long before the advent of virtual reality, the most sophisticated form of immersive media was the "armchair travel" stereoscopic box set [Darrah 1977]. Pioneered by publishers such as Underwood & Underwood, these were not mere collections of images but carefully curated multimedia systems designed to simulate a virtual tour to distant lands. A typical set included a handheld viewer, up to 100 numbered stereographs, detailed maps charting the location of each photograph, and an expert guidebook. The technical core of this experience was the stereograph itself: a card with two photographs of the same scene captured from slightly different perspectives. When seen through a stereoscope, the brain fuses these paired images into a single view, creating a convincing and often startling illusion of three-dimensional depth.

For a public to whom international travel was largely inaccessible, these sets provided a powerful and educational "virtual tour" that fundamentally shaped how Western audiences perceived distant cultures and events. The combination of the immersive 3D illusion with a structured, first-person narrative allowed users to do more than just view images; they could follow a curated journey, witness historical moments, and even confront the battlefields of World War I, which were rigorously documented in this format [Darrah 1977]. This power and accessibility made the stereoscope a ubiquitous fixture in households and the dominant visual mass medium of its era.

Today, this powerful photographic medium has been largely forgotten by the public, though an estimated seven million stereographs were produced [Darrah 1977], and many complete box sets survive. These archives offer invaluable immersive windows into the past, and could allow us to revisit early paradigms of presence, virtuality, and the curated experience. Increasingly ubiquitous modern virtual reality devices are perfectly suited to reconstituting the stereo illusion and revive these collections for contemporary



Figure 1: Antique stereographs and a foldable stereoscope. Photograph courtesy of the authors.

audiences. However, simply digitizing and displaying these collections is insufficient. The curated tours they present are products of their time, with narratives deeply entangled with the imperialistic and civilizationist ideologies of the era. Reviving this material for modern audiences therefore requires a critical framework that can deconstruct the original narrative logic without censoring the challenging perspectives contained within.

While several projects have explored the visualization of individual stereographs in VR [Boehlert 2016; Ricupero et al. 2019; Taipina and Cardoso 2022], to our knowledge none have attempted to revive the box set as a complete narrative system. This paper presents such a framework, leveraging Natural Language Processing methods to reconfigure the relationships between text and image to create new digital representations suitable for exhibition in virtual reality. We demonstrate this framework through two artworks that reinterpret a particularly compelling case study: James Ricalton’s *China Through the Stereoscope*

2 The Archive: Ricalton’s China in 3D

2.1 Content and Ideological Context

James Ricalton’s *China Through the Stereoscope* was published in 1901 by Underwood & Underwood. It begins as a typical stereo travelogue discussing Chinese commerce, daily life, architecture, local customs, and natural resources. However, Ricalton’s itinerary was abruptly interrupted by the Boxer Uprising, a violent anti-foreign and anti-Christian rebellion, thrusting the reader into the midst of the conflict. As both photographer and author, Ricalton acts as a war correspondent and personal guide. His 100 stereographs transcend typical travel scenery, showing viewers the ‘horrors of war’: scenes of destruction, dead soldiers, ‘Boxer’ prisoners, and gruesome warnings erected by occupying forces, such as the heads of beheaded criminals. Ricalton’s accompanying book framed the experience as a guided tour, with 100 chapters each contextualizing one of the stereographs. It positioned the stereoscope as a “miracle of realism” that allowed audiences to feel as though they were truly

witnessing events they had only read about in newspapers [Ricalton 1901].

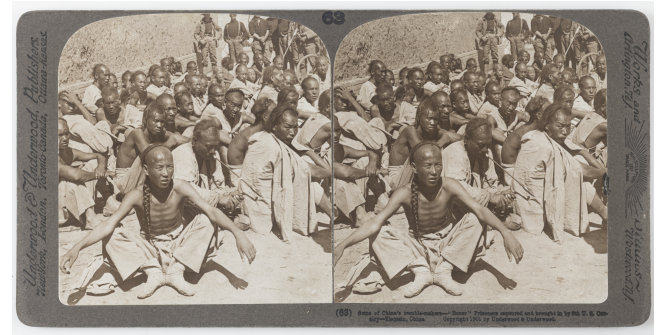


Figure 2: "Some of China’s troublemakers-"Boxer" prisoners captured and brought in by 6th U.S. Cavalry - Tientain, China", 1901. Image 63/100. Courtesy of Moonchu Foundation

Yet, Ricalton’s work is a product of its time, and his commentary frequently reveals a perspective deeply entangled with the imperialistic and civilizationist ideologies of the era. His practice of “stereoscopic ethnography” consistently frames his subjects through a lens of American exceptionalism and social Darwinism [Winter 2018]. Such perspectives, which stand out to modern audiences, are representative of most other stereo commentaries of the time. Reviving and exhibiting this material requires acknowledging and critically addressing (without censoring or altering) such perspectives for modern audiences, which in itself represents an interesting scholarly exercise.

2.2 From Artifact to Dataset: Digitization and Preparation

2.2.1 The Book. A scan of the book was acquired from the UCLA California Public Library via the Internet Archive. Existing OCR of the source guidebook was unreliable. We generated a clean text corpus for NLP analysis by re-transcribing the book using Google Cloud’s Vision API. The text was then subdivided into chapters, paragraphs, and individual sentences, forming the textual basis of our dataset.

2.2.2 The Stereographs. A complete 100-stereograph set generously donated by the Moonchu Foundation was digitized with a medium-format camera at 3500x7000 resolution. A restoration workflow, including cropping, alignment, and damage repair, produced artifact-free image pairs suitable for comfortable stereoscopic display.

3 Methodology

Our artworks require transforming and challenging the linear narrative of the book and its accompanying images to present the corpus to a modern, non-expert audience. This was done using computational and AI-assisted methods to extract, process, analyze, and reframe its different elements.

3.1 Reconfiguring text structures through LLMs

To create an alternative to the book’s chronological organization, we implemented the topic modeling framework of van Wanrooij et al. [van Wanrooij et al. 2024]. The corpus was segmented into paragraphs as optimal thematic units, with an LLM generating a comprehensive topic list and assigning a single topic per paragraph. To enable combinatorial queries (e.g., ‘Food’ + ‘War’ = nutrition during wartime), we developed a multi-label sentence-level assignment system leveraging LLM response probabilities: For each sentence-topic pair, the model determined relevance while logarithmic probabilities quantified assignment confidence, thereby establishing a probabilistic framework for polythematic categorization.

This methodology constitutes Cross Eyed’s core interaction model and represents a deliberate curatorial shift. Where Ricalton imposed a linear narrative, our LLM collaboration exposes latent thematic networks within his text, enabling polyvocal interpretations that transcend authorial intent.

3.2 Exploring the world of text embeddings

A different approach for deconstructing the text employs semantic embeddings via the BGE-M3 model, encoding textual elements into a 1024-dimensional latent space to quantify content relationships through vector operations [Chen et al. 2024]. User queries are similarly embedded, enabling semantic distance calculations for content retrieval and relevance visualization [Reimers and Gurevych 2019]. Under the premise that chapter text fragments intrinsically link to their corresponding images, an image ranking strategy is implemented: The 50 nearest text embeddings to a query (empirically determined) are normalized, inverted, and aggregated into a composite score reflecting proximity and frequency. This facilitates image search functionality. For spatial representation, Uniform Manifold Approximation and Projection (UMAP) reduces dimensionality to three axes while preserving global thematic structure [McInnes et al. 2020].

Critically, this semantic landscape constitutes a cultural artifact shaped by the BGE-M3 model’s architecture and training data. The resulting projection is not an objective textual representation but a technologically mediated interpretation. Latent Cartographies invites users to navigate this construct, simultaneously exposing Ricalton’s original cartography and its AI-generated counterpart.

3.3 Automatic Decolonization

As established in the Introduction, this corpus features colonial language requiring systematic detection and contextualization. To identify optimal automated detection methods, we evaluated multiple techniques against a manually curated validation set of 140 sentences (70 offensive, 70 neutral). Two primary approaches were assessed: sentiment analysis models detecting affective polarity, and an LLM-based classifier inspired by Chiu et al. [Chiu et al. 2021]. The latter was deployed in two configurations - with and without historical contextualization noting the text’s 1900 Sino-centric provenance.

Performance results (Table 1) demonstrate that while no method achieves perfection, the contextualized LLM approach significantly outperforms sentiment analysis. This detector is implemented in

the Cross Eyed installation for real-time textual flagging. Critically, these annotations function not as definitive judgments but as epistemic provocations: By rendering the AI’s assessments visible, the work foregrounds the inherent challenges in operationalizing colonial discourse detection, directing attention to the constitutive ambiguities of such language itself.

Table 1: Colonial Language Detection Methods

Model	F1-Score	Comments
NLTK VADER	0.477	Sentiment Analysis
TextBlob	0.367	Sentiment Analysis
Transformers	0.537	Sentiment Analysis
LLM-based	0.816	w/-out context
LLM-based	0.867	w/ context

4 A Pair of Artworks

We presented two artworks borne from our computational reconfiguration of the archive. Both installations reframe the modern user’s engagement with a historical stereograph box set, to reinterpret this narrative journey through 20th century China, challenging the colonial and imperialist discourse of the time.

Both artworks are presented in an 8x4m, cylindrical VR multi-user theatre based on the AVIE system [McGinity et al. 2007], using active shutter glasses, a 29.4-channel spatial audio system, and HTC Vive-tracked tablets for interaction. This stereoscopic system reconstitutes the stereo illusion of Ricalton’s stereographs, allowing them to be viewed as intended: in 3D.

4.1 Cross Eyed: Decolonizing the Gaze of 19th Century Stereophotography

Cross Eyed reimagines engagement with the historical archive by inviting users to view it through a series of conceptual lenses. Its dual aim is to make the dense material more accessible while creating a visual paradigm for critically decolonizing the author’s gaze.

These lenses (such as ‘Colonialism’, ‘Missionary Work’, or ‘Family Structures’) are thematic categories generated by the LLM-based topic modeling strategy detailed in Section 3.1. Via a tablet interface, visitors can select a single lens or combine multiple lenses (e.g., ‘War’ and ‘Food’) to create combinatorial queries. This interaction model bypasses the book’s linear narrative, instead revealing the latent, often unexpected, connections between different subjects in Ricalton’s journey, and gives users accessible entry points to query the archive.

Upon selection, the 360-degree screen is populated by corresponding stereographs and scrolling text fragments. The artwork’s visual language is built around the ticker tape, a medium chosen for both its potent psychological effect and its rich cultural meaning. The primary goal is to compel engagement with historical text that viewers might otherwise ignore by inducing a state of ‘FOMO’. Drawing on the ticker’s innate affective properties, we leverage two fundamental cognitive principles. First, the continuous horizontal



Figure 3: Visitors viewing stereographs and commentaries through the 'War' lens within *Cross Eyed*.

motion creates strong visual salience, involuntarily capturing attention in a way static text cannot. Second, the appearance of the beginning of a message creates the need to see the end, and the ticker satisfies that need, but only by introducing the beginning of the next message, thus renewing the cycle. This keeps the viewer "hooked" in a way that fully-presented static text cannot, effectively "hooking" them into a state of sustained reading.

Beyond this cognitive hook, our choice of the ticker is a critical intervention inspired by artists like Jenny Holzer, who subverted the medium's authority. The ticker is culturally coded; we associate its form with the objective, real-time authority of financial data and breaking news. By channeling Ricalton's subjective, often biased, 100-year-old voice through this modern medium of "truth," we create a deliberate cognitive dissonance. The artwork casts Ricalton as a contemporary war correspondent, but the clash between the authoritative format and the historically-contingent, imperialist content forces a critical re-evaluation. The viewer is no longer passively consuming a historical text; they are actively grappling with the mismatch between the medium and its message, which is the central goal of decolonizing Ricalton's gaze.

To make this dissonance more explicit, our system uses the algorithm developed in Section 3.3 to detect and visually flag sentences containing imperialist or racist perspectives. The text is not censored; instead, problematic passages are subtly distinguished from neutral descriptions by their color and typography. This visual distinction transforms the act of reading into an act of critical inquiry, empowering viewers to identify the author's biases and form their own judgments about a historical remnant presented in its entirety.

4.2 Latent Cartographies

Diverging from the topic-centric framework of the first artwork, this piece positions the continuous latent space as the foundation of an interaction paradigm prioritizing user agency over algorithmic suggestion. Inspired by the author's foreword on physical and conscious realities, we developed two interdependent scenarios:

Conscious Realm: Users entering the cylindrical display encounter a three-dimensional semantic landscape where text fragments are spatially organized by UMAP-reduced text embeddings. Initially imperceptible, the latent structure manifests as a seemingly chaotic constellation. Sparse clusters of free-floating phrases within an unbounded space evoke cognitive associations, situating spectators within the author's mental universe. This reorganization enables

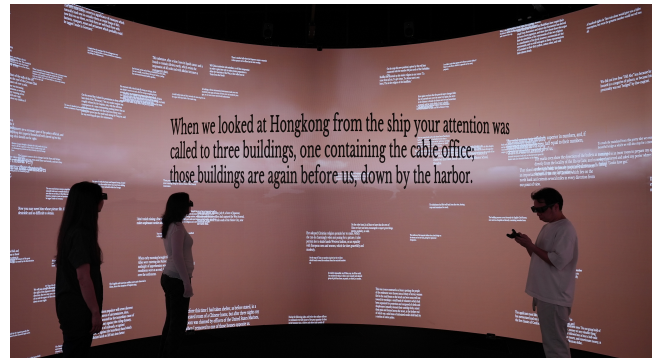


Figure 4: Visitors engaging with the Conscious Realm within the artwork *Latent Cartographies*

novel exploration of semantically linked passages, circumventing both the original structure and authorial intent. Prompting the system either by text or speech input, propels the audience to new coordinates within this space. As the text slowly passes their gaze, visitors may catch inspiration for their next prompt. Upon arrival, the most relevant text fragment centers in front of them (Figure 4). While cursor-based navigation remains possible, its functionality is deliberately constrained to privilege natural language interaction. To further deepen immersion, manually positioned harmonic phrases from piano, violin, and cello inhabit the same 3D space. This ambient sonification amplifies the illusion of infinite expanse while providing navigational cues during latent space traversal.



Figure 5: Visitors exploring the Physical Realm within the artwork *Latent Cartographies*

Physical Realm: Selecting a text fragment transitions users to a cylindrical arrangement of the complete image collection, with the corresponding image foregrounded (though not yet selected). Cursor hovering dynamically modulates image brightness and spatial positioning: targeted images advance toward the viewer while laterally displacing adjacent content. Users may also employ natural language queries to rank images by semantic proximity to the query, as described in Chapter 3.2. Selecting an image moves a topographical map of China in the background to the chapter's region, reveals the photo's precise location on an accompanied map and displays the full chapter. Text fragments are highlighted based

on similarity to the user's query history, guiding exploration of thematic regions. Users may annotate passages as "concerning," fostering immediate critical discourse and centering human interpretation in meaning-making. Clicking any text fragment returns the user to the Conscious Realm, revealing its semantic cluster and completing the interaction loop.

5 Conclusion

This research contributes a dual computational framework for the critical reconstitution of colonial-era stereoscopic box sets demonstrated through two VR artworks that offer contrasting modes of engagement, using James Ricalton's *China through the Stereoscope* as our case study. The first, Cross Eyed, embodies a top-down, analytical approach. It uses LLM-generated thematic lenses to systematically deconstruct the archive, enable access through the new mediated structure, and visually highlight imperialistic rhetoric through visual interventions. The second, Latent Cartographies, employs a bottom-up, exploratory paradigm. It transforms the archive into an emergent semantic landscape, empowering users to navigate the latent connections between text and image through their own eager inquiry.

Despite their divergent strategies, both artworks achieve the shared critical objective of decentralizing the author's singular narrative. They transform passive spectators into active participants who must grapple with historical bias as a tangible, spatial phenomenon within the 360° immersive environment. This methodology reframes archival interrogation as an experiential process rather than a didactic lesson, using the sensory immediacy of VR to complicate the viewer's relationship to the past, but also facilitate access to it through reconstituted stereoscopic photographs.

As a model for critical media practice, this project demonstrates how machine learning can be engaged not as a neutral tool for presentation, but as a collaborative partner in artistic and historical interpretation. By creating a living archive where past and present continuously reconfigure each other, our framework offers new pathways for examining the technological infrastructures of representation. It is an extensible model that could be applied to other contested collections, fostering restorative dialogues with difficult heritage. Evaluations are planned to gauge and refine the effectiveness of our approach in the future.

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